



**SABARMATI DIESELS, LOCO CARE CENTRE, SABARMATI
ELECTRICAL**

[3फेज विद्युत लोकोमोटिव आई ए/आई बी/आई सी अनुसूची [आवक/जावक निरीक्षण]
IA/IB/IC Schedule of 3Phase Electric Locomotive [Incoming/Out Going Inspection]

पश्चिम रेलवे
Western Railway

लोको संख्या Loco No 30502 क्लास Class WAP 7 शिड्यूल में लेने का दिनांक Date taken under Sch. 16/04/26

	आगमन Incoming / शिड्यूल Schedule		अंतिम निरीक्षण Final Inspection	
	अपर डैक Upper Deck	अंडर ट्रक Under Truck	अपर डैक Upper Deck	अंडर ट्रक Under Truck
Date/ Shift	16/04/26 / 08/17	16/04/26 / 08/17	16.4/26 18/04/26	18.4/26
Name of Technician	PAPPU GUPTA	SANTOSH KUMAR	Pooja	Shreshth
Designation/ Token No.	ELF- Ist	ELF- Ist	ELF- Ist	ELF- Ist
Signature	Pappu Gupta	Santosh Kumar	Pooja	Shreshth

Customer Feed Back/Bookings (Record feedback given by Loco pilot)				
Sr. No.	Customer Feed Back/Bookings	Action Taken	Name of TCN / Attended By	Remarks, if any & Sign. of JE/SSE
①	RTIS Display Broken Not working.	Same Allowed	—	See. J.
②	SR#2 Coolant कम है।	Filled.	—	See. J.
③	MRB#2 Abnormal Sound आ रहा है।	MRB#2 changed by BH	—	See. J.
④	TM#1 connection loose खूली है।	Clean tight	Arund	See. J.

Cab AC not available.

ज. इंजी/सी.से.इंजी का हस्ताक्षर Signature of JE/SSE

21/4

INSPECTION WING ELECTRICAL: CHECK AND RECORD FOLLOWINGS

क्र.सं SN	कार्यवाही कार्य /विवरण का निरीक्षण / (शिड्यूल) Detail of Work/ Inspection (Schedule)			कार्यवाही की गयी Action Taken		
A	आगमन INCOMING (WHEN LOCO IS ENERGISED)					
जॉब की सूची CHECK LIST						
	विवरण Description	मानक मान Standard Value	वास्तविक मान Actual Value	Cab-1	Cab-2	हस्ताक्षर टिप्पणी Sign/ Remarks
1	Battery Voltage	100-110 V		99V	101V	
2	CHBA Voltage	95-110 V		110V	110V	
3	Charging Current	10 Amp		6A		
4	Working in cooling mode both Cab	Working		Not working.		
5	Working of Flasher light, Marker light, Cab light, Signaling light, Indication lamp, Instrument light, corridor light, Cab fan, Cab AC, Cab Heater in both cabs.	Working		working	working	
6	Working of meter UBA, U& TE/BE (Bogie-1); TE/BE (Bogie-2) by both Cab	Working		working	working	
7	Working of head light contactor CPR & head light (Full/Dimmer/Rear). Check focus of Head light	Working		working	working	
8	Check Regression in both cab by A-9 and RS	Checked		check ok		
9	RGCP: Opens at (cut out in kg/cm ²) Close at (cut in kg/cm ²)	10.0 ± 0.20 8.0 ± 0.20		10.0 kg/cm ² 8.1 kg/cm ²		
10	Working of cooling fan (Churney fan) BUR-1, BUR-2, BUR-3, BUR-4	Working		OK		
11	Working of cooling fan SR-1, SR-2, CEL-1, CEL-2	Working		OK		
12	Condition of Silica Gel of SR-1, SR-2	Checked		check ok		
13	Check Auto Flasher light by RS only and its switches.	Working		working	working	
14	Working of Flasher light by BPFL (Flasher lights are working in main & standby.)	Working		working	working	
15	Check Working of ACP, Check Working of PVEF	Working		working	working	
16	Check and Ensure proper functioning of vigilance control device	Working		working	working	
17	FDU unit set voltage at test point	0 ± 50 mv	-147 mv (set - 30mv)			
18	Working of FDU by smoke to be checked the only one cab	Working	working.			
19	Working of monitor (DDS) Both cab	Working		working	working	
20	Match the OHE meter with DDS screen (should be same)	Same		Same	Same	
21	Match the time of SPM with DDS screen (should be same)	Same		Same	Same	
22	Check the working of the Emergency Push Button Switch for correct operation.	Checked		check ok	check ok	


 ज. इंजी/सी.से.इंजी का हस्ताक्षर Signature of JE/SSE
 16.04.26

कार्यवाही कार्य / विवरण का निरीक्षण / (शिफ्ट्यूल) Detail of Work/ Inspection (Schedule)					मानक मान Standard Value	वास्तविक मान Actual Value		हस्ताक्षर/ टिप्पणी Sign/Remark
						Cab-1	Cab-2	
Checking Brake Power (Loco should not move on 150 KN of TE)					150 KN			
24	Check millivolt drop MV-218 Contactor (SMI no. 275)				< 50mV	9.0 mV		
25	Check millivolt drop MV-126 Contactor (SMI no. 275)				< 50mV	19.9 mV		
26	Status of sub system from SS-01 to SS-19 (It should be 00)				All are 00	All Healthy		
27	Check all foot switches operation. Connection and connection terminal not to touching with cover.				Checked	Check ok		
28	Check oil level of SR should be in.		SR-1		Between Min & Max	Between		
	Check oil level of SR should be in.		SR-2		Between Min & Max	Between		
29	Check Earth return current (P5/P7)							
	Axle no.	Axle no.	Axle no.	Axle no.				
	0.3	2.5	1.8	1.1				
30	Function test :							
	Test function Earth Fault							
	Control ckt Positive		Positive PCLH + Earth (89.7)					
	Negative		Negative PCLH + Earth		check ok			
	Harmonic Filter		Earth Link (89.6)		-DC-			
	Aux. Ckt.		Earth + 1117 IN HB2 (89.2)		check ok			
	415/110		HB1 Earth + 1218 (89.5)		check ok			
31	Performance of BURs when one BUR is isolated				BUR-1	BUR-2	BUR-3	
	When any one BUR goes out then rest of the two BUR's should take the load of all the auxiliaries at ventilation level 3 of the loco. OK/Not OK				check ok	check ok	check ok	
32	Temp. Value of TM as seen in Driver Display.							
	TM1	TM2	TM3	TM4	TM5	TM6		
	35	36	36	35	36	36		
33	Take the traction both in forward & reverse direction and ensure 596 nodes appear on screen.		मानक मान		Cab-1	Cab-2		
			Yes / No		yes	yes		
34	Instruction stickers in cab provided		Yes / No		yes	yes		
B	GAUGES AND PRESSURE SWITCHES							
a)	Ensure proper position of gauges and no difference in both cab's gauge, Check all gauge light working.				check ok			
b)	Check and ensure zero error and no leakage in connection.				check ok			
c)	Check operation of corridor tube light and its switch.				check ok			
d)	Check fitment and operation of spot light bulb and holder in both Cab				check ok			
e)	Check all earthing points located on equipment.				check ok			

ज. इंजी/सी.से.इंजी का हस्ताक्षर
Signature of JE/SSE
16.04.26

क्र.सं.	कार्यवाही कार्यविवरण का निरीक्षण // (शिक्यूल) Detail of Work/ Inspection) Schedule)	मानक मान Standard Value	वास्तविक मान Actual Value																																																	
C	Rotating Machines (MRB-1&2, SCMRB 1&2, SCTM 1&2, OCB 1&2, TMB 1&2, CP-1 &2, CPA), SR & TFP Pump																																																			
	While Loco is energised, check the following:																																																			
a)	Check abnormal sound, temperature & sparking, Oil leakages and proper functioning of all auxiliary motors/Pumps.	check ok																																																		
b)	Ensure that the electrical machines and its cables are adequately protected & covered thus preventing ingress of oil	check ok																																																		
c)	Check direction if any motor changed.																																																			
d)	Measure the current of all four oil pumps. (IC Sch.) Measure the current of all auxiliary motors																																																			
	<table border="1"> <thead> <tr> <th rowspan="2">MCB</th> <th colspan="2">HB#1</th> <th colspan="2">HB#2</th> </tr> <tr> <th>Starting</th> <th>Stable</th> <th>Starting</th> <th>Stable</th> </tr> </thead> <tbody> <tr> <td>62.1</td> <td>19.4</td> <td>12.3</td> <td>18.4</td> <td>11.4</td> </tr> <tr> <td>63.1</td> <td>17.8</td> <td>4.8</td> <td>18.2</td> <td>4.9</td> </tr> <tr> <td>47.1</td> <td>13.2</td> <td>26.8</td> <td>13.9</td> <td>26.4</td> </tr> <tr> <td>53.1</td> <td>11.9</td> <td>18.4</td> <td>13.1</td> <td>22.2</td> </tr> <tr> <td>55.1</td> <td>22</td> <td>4.9</td> <td>27</td> <td>5.4</td> </tr> <tr> <td>59.1</td> <td>15.7</td> <td>26.2</td> <td>14.3</td> <td>25.8</td> </tr> <tr> <td>54.1</td> <td>24.4</td> <td>5.8</td> <td>28.2</td> <td>6.0</td> </tr> <tr> <td>56.1</td> <td>19.2</td> <td>3.2</td> <td>17.1</td> <td>2.8</td> </tr> </tbody> </table>	MCB	HB#1		HB#2		Starting	Stable	Starting	Stable	62.1	19.4	12.3	18.4	11.4	63.1	17.8	4.8	18.2	4.9	47.1	13.2	26.8	13.9	26.4	53.1	11.9	18.4	13.1	22.2	55.1	22	4.9	27	5.4	59.1	15.7	26.2	14.3	25.8	54.1	24.4	5.8	28.2	6.0	56.1	19.2	3.2	17.1	2.8		
MCB	HB#1		HB#2																																																	
	Starting	Stable	Starting	Stable																																																
62.1	19.4	12.3	18.4	11.4																																																
63.1	17.8	4.8	18.2	4.9																																																
47.1	13.2	26.8	13.9	26.4																																																
53.1	11.9	18.4	13.1	22.2																																																
55.1	22	4.9	27	5.4																																																
59.1	15.7	26.2	14.3	25.8																																																
54.1	24.4	5.8	28.2	6.0																																																
56.1	19.2	3.2	17.1	2.8																																																
D	While Loco is energised, Check and attend for proper working of																																																			
	Working of all the rotating switches in SB-1 panel to be checked (should be in normal position)																																																			
	<table border="1"> <thead> <tr> <th>Switch no.</th> <th>Position</th> <th>Switch no.</th> <th>Position</th> </tr> </thead> <tbody> <tr> <td>154</td> <td>Normal</td> <td>160</td> <td>'1'</td> </tr> <tr> <td>152</td> <td>'0'</td> <td>237.1</td> <td>'1'</td> </tr> </tbody> </table>	Switch no.	Position	Switch no.	Position	154	Normal	160	'1'	152	'0'	237.1	'1'	check ok																																						
Switch no.	Position	Switch no.	Position																																																	
154	Normal	160	'1'																																																	
152	'0'	237.1	'1'																																																	
E	Air delivery measurement in 3-phase locomotive to ascertain proper cooling and pressurization of Machine Room. (As per RDSO/2009/EL/SMI/0255 Rev.'0', dtd.08.05.2009)																																																			
During IRC work																																																				
	SCTMB & OCB#1,2 (Min:13m/s)	MRB#1&2 (Min:03m/s)	SCMRB#1&2 (Min:12m/s)	OCB#1 (Min: 08 m/s)	OCB#2 (Min: 08 m/s)	TMB#1&2 (Min:12m/s)	हस्ताक्षर																																													
1	29.8	13.5	18.9	10.8 15.1 13.8	12.8 12.1 15.1	15.8	Ramesh																																													
				10.1 10.5 10.6	14.1 15.1 9.3	14.9																																														
				10.8 11.8 10.1	10.1 10.3 13.8	16.0																																														
2	27.5	10.2	20.0	Air flow meter not working		15.1																																														
						15.1																																														
During FRC work																																																				
	SCTMB & OCB#1,2 (Min:13m/s)	MRB#1&2 (Min:03m/s)	SCMRB#1&2 (Min:12m/s)	OCB#1 (Min: 08 m/s)	OCB#2 (Min: 08 m/s)	TMB#1&2 (Min:12m/s)	हस्ताक्षर																																													
1	29.8	13.5	18.9	10.8 15.1 13.8	12.8 12.1 15.1	15.8	Ramesh																																													
				10.1 10.5 10.6	14.1 15.1 9.3	14.9																																														
				10.8 11.8 10.1	10.1 10.3 13.8	16.0																																														
2	27.5	10.2	20.0	10.8 11.8 10.1	10.1 10.3 13.8	15.1																																														
						15.1																																														
Check Earthing shunt of Bogie to Body																																																				
	Axle box 1-3	OK	Axle box 2-4	OK																																																
	Axle box 5-7	OK	Axle box 6-8	OK																																																
Check Earthing shunt of Axle to Bogie																																																				
	Axle box 2	OK	Axle box 3	OK																																																
	Axle box 6	OK	Axle box 7	OK																																																
F	CVVRS: Check proper function and setting of all cameras. Ensure date and time also.																																																			

ज. इंजी/सी.से.इंजी का हस्ताक्षर Signature of JE/SSE

कार्यवाही कार्यविवरण का निरीक्षण // (शिफ्ट)	मानक मान Standard Value	वास्तविक मान Actual Value	हस्ताक्षर/ टिप्पणी Sign/Rem						
ENERGY CUM SPEED MONITORING SYSTEM (ESMON)									
a) Visually check the ESMON for any abnormality and send the memory card to section for data retrieving.									
b) Energy Consumed		22808588 KWH							
c) Energy Regenerated		3539975.7 KWH							
d) Total KM		59325 KM							
e) Working of Speedometer		Working.							
H Simulation mode:									
a) Vigilance working from both cabs		Working.							
b) 160 kept on 0 ensure speed showing 14 kmph		Working.							
c) Check both SR Contactor air leakage 12.4/1 & 12.4/2		—							
d) FB Cubical contactor air leakage (8.1 & 8.2)		—							
I Check operation of Vigilance Control Device (VCD)									
VCD becomes active when any one or more operation listed below is not performed for continuous 60 seconds (1 minute)		Check OK							
i) Change of TE	ii) Application of PVCD								
iii) Sander operation	iv) Application of brake A-9								
v) Application of brake SA-9									
Check following LED Indication/Status of following items after elapse of 60 sec									
SN	Indication	Status of Indication							
		From 60 to 68±2 sec		From 68±2 to 76 sec		After 76 sec		After 76 + 34 sec	
		Std	Observed	Std	Observed	Std	Observed	Std	Observed
1	LED Indication	ON	OK	ON	OK	ON	OK	OFF	OK
2	Buzzer sound	OFF	OK	ON	OK	OFF	OK	OFF	OK
3	VCD Pneumatic Valve	OFF	OK	OFF	OK	ON	OK	ON	OK
4	Penalty brake	OFF	OK	OFF	OK	ON	OK	ON	OK
5	Re-setting of VCD by any one or more of the above operations	YES	OK	YES	OK	NO	OK	Only by re-set switch	OK
J ERROR LOG									
Fault code	Error log	Action Taken							

अन्य किसी भी जानकारी Any Other Information:

Sachin
16.04.24
Sachin Bisla 8/17
Signature and Name of JE/SSE
Incoming Inspection with date & shift

Signature
Signature and Name of JE/SSE for
Final Inspection with date & shift

SCHEDULE PROGRESS CHART

SN	Date	Shift	QC	Work Details	Name of JE/SSE	Sign of JE/SSE	Remark
1	16.04.26	8/17	EL	IRC Sch Done	Sachin B.	Sachin B.	
2	17.4.26	8/12	EL	Cab 21 Sch done			Rajesh P. P.
3	"	"	"	Ann Machine Sch & Scaffolding			R.P. + Ashwini
4	"	"	"	On 1st & 2nd station done	Sachin B.		Ann + N.K. + Ashwini
5	"	16/12	"	Cab 22 Sch done	Bell & S. K.		S.K. + M. K.
6							
7							
8	17/4/26	8/12	EL	SR & Box Sch done			
9				By medha staff			medha staff
10				Cab Heater & Kavach			
11				Sch done			
12	18/4/26	90	EL	Both Cab Heater Sch Done	Kantilal		Grishay + medha
13	19/4/26	18pm	"	On 1st station & fire done	Bell & S. K.		Rajesh P. P.
14							
15							
16							
17							
18							
19							
20							
21							
22							
23							
24							
25							



S. K.

मी. से. इजी इंजार्ज का हस्ताक्षर Signature of SSE In-charge

S. K.

**LOCOMOTIVE SHED, SABARMATI
ELECTRICAL**

**[3-फेज विद्युत लोकोमोटिव आईए/आईबी/आईसी अनुसूची [निरीक्षण]
IA/IB/IC Schedule of 3-Phase Electric Locomotive [Inspection]**



पश्चिम रेलवे
Western Railway



Cab #1

लोको संख्या Loco No. 30502 शिड्यूल Sch. IB शिड्यूल में लेने का दिनांक Date taken under Sch. 17-4-20

क्र.सं. SN	कार्यवाही कार्य / निरीक्षण का विवरण / (शिड्यूल) Detail of Work/ Inspection (Schedule)	कार्रवाई की गयी Action Taken	टीसीएन का नाम Name of TCN	हस्ताक्षर / टिप्पणी Sign / Remarks
A	डाइविंग डेस्क (Driving Desk)			
1	Clean the surfaces of the driver's desk with a soap solution and cloth, taking care not to allow water to enter electrical cubicles.	OK		
2	Clean the dust filter on the buzzer using a soft brush without using the pressure / compressed air.	OK		
3	Ensure that transparent rubber caps provided over the push buttons of driver desk.	OK		
4	Clean the battery, OHE and TE/BE meters in both cabs and check for zero error	OK		
5	Check tightness/operation/cable rubbing of connector, earthing of angle transmitter, all rubber gasket and all equipment located on BL and its box frame inside the panel. Clean contacts, check cap and its fitment tightness.	OK		
6	Check tightness of sub D located on BL.	OK		
7	Check ZPT switch located on pneumatic panel.	OK		
8	Check all earthing points located on equipment.	OK		
	LIGHTING AND AIRFLOW			
B	हेड लाइट, कैब व कोरिडोर लाइट, फ्लैशर लाइट, मार्कर लाइट HEAD LIGHT, CAB & CORIDOOOR LIGHT, FLASHER LIGHT, MARKER LIGHT			
1	Check lighting units for physical damage, tightness of fixing connections, particularly look for signs of overheating or burning. Replace bulbs/tubes, which have failed.	OK		
2	Clean all internal light unit diffusers, bulbs or tubes and reflectors.	OK		
3	Clean the glasses of the headlight lamp and marker light.	OK		
4	Ensure that there is no entry of water through head light lamps.	OK		
5	Check for crack or breakage of the aluminum casing of headlight beams.	OK		
6	Check the proper operation of flasher light, marker light (red & white), headlight, cab lights, disc light, m/c. room light.	OK		
7	holder and clean glass Check headlight & reflector. check tightness of headlight convertor, Ensure sealing	OK		
8	Clean marker light glass and check tightness of connection. Check looseness, flashing in flasher unit.	OK		

रमेश पटेल

क्र.सं. SN	कार्यवाही कार्य/निरीक्षण का विवरण / (शिड्यूल) Detail of Work/ Inspection (Schedule)	कार्रवाई की गयी Action Taken	टीसीएन का नाम Name of TCN	हस्ताक्षर / टिप्पणी Sign / Remarks
C	SB1 & 2, HB1 & 2 AND FB PANEL / CUBICLE			
1	Check the mounting block of 8.1 and 8.2 contactors by opening side windows and check the main contactor, pre charging contactor foundation bolt tips for looseness and overheating by removing arc-chutes from front side.	—		
2	Check the condition of capacitors for leakage and check earth fault relay, resistance and discharging resistance for overheating. Check the tightness of all fastener lug, LEM sensor and CTs.	—		
3	Ensure that the various foundations bolts, body support bolts and supports plates of FB cubicle are intact and tight. Check tightness of cubical doors safety locks.	—		
4	Ensure tightness of various electrical connections, CBS in SB, HB cubicles.	OK		
5	Check back side of HB cubicle all SB connections for tightness/overheating/cable rubbing.	OK		
6	IC Sch.: a) Clean various equipment fitted inside the SB, HB & FB cubicle.	—		
	b) Visual examination of various EPCs & EMCs fitted inside FB cubicle and ensure that there is no crackness of cover cylinder, leakage and they are functioning properly.	—		
	c) Check the tightness of various PCB cards in central electronics.	—		
	d) Check the RC input filter circuit for any abnormality and measure the resistance value.	—		
	e) Take out the arc-chutes of pre-charging contactors for looseness of strips and ensure tightness of its armature locking nut by hand.	—		
	f) Check the FB cable junction for tightness, cable rubbing and overheating.	—		
D	HB CUBICLE :			
1	Check all equipments' fitment, tightness, intactness of base foundation bolts & body support.	OK		
2	Ensure earthing point/shunts of all panels/cubicles.	OK		
3	Check all MCB cable connections & Siemen connector tightness and no overheating.	OK		
4	Check tightness of 80 A & 150 A contactors connection and condition of tips.	OK		
5	Check 1000 V/415 V/110 V transformer connections for overheating and tightness of foundation bolts and checking of fuses for healthy connection.	OK		
6	Check tightness and condition of capacitor, LEM sensor below HB panel, earth fault relay & resistance connection, RLC plate, temperature sensor and all diode boxes.	OK		
7	Checking of OCB cable in HB 1& 2 for overheating cable no.1121, 1122, 1123 (A & B).	OK		

कार्यवाही
Detail

क्र.सं.
SN

कार्यवाही
Detail

कार्यवाही
Detail

Handwritten notes in right margin:

Handwritten signature/initials.

क्र.सं. SN	कार्यवाही कार्य / निरीक्षण का विवरण / (शिड्यूल) Detail of Work/ Inspection (Schedule)	कार्रवाई की गयी Action Taken	टीसीएन का नाम Name of TCN	हस्ताक्षर / टिप्पणी Sign / Remarks
F	SB CUBICLE:			
1	Check fasten lug connected on various equipment. Fitment, tightness	OK		
2	Check intactness of base foundation bolts & body support.	OK		
3	Check tightness of cubical doors safety locks.	OK		
3	Check all EMC connector lug position layout. It should not be touch with plunger.	OK		
4	Visually check the connections on backside of SB 1 & 2 panel for overheating of WAGO connector etc.	OK		
5	Check tightness of DC-DC converter 24V 48V and Siemen connector for overheating.	OK		
6	Visually check all pressure switches, tightness of MCB, MCR relay, no voltage relay connections.	OK		
7	Key interlocking: Check all keys are present & the complete system is operational.	OK		
8	Ensure the healthiness: Visually check all pre/proper functioning of rotating switch 152, 154 and 237.1 on its various locations.	OK		
9	Check visually for any abnormality in SB, HB & FB cubicle	OK		
10	Ensure proper functioning of thermostat of central electronics. (IC Sch.)	—		
F	MAIN ASSEMBLY BOX-1 & BOX-2:			
1	Clean the dust of enclosure using vacuum cleaner.	OK		
2	Check the metallic strip provided for spring action in the contact is intact. Replace the contact if required.	OK		
3	Check the connections for 25, 50 & 70 Sq. mm cables for their tightness. (IC Sch.)	—		
4	Tightness of knife contact connection (female), 3-phase choke connection, Power contactor connection. (IC Sch.)	—		

क्र.सं. SN	कार्यवाही कार्य / निरीक्षण का विवरण / (शिड्यूल) Detail of Work/ Inspection (Schedule)	कार्रवाई की गयी Action Taken	टीसीएन का नाम Name of TCN	हस्ताक्षर / टिप्पणी Sign / Remarks
G	CENTRAL ELECTRONIC RACK			
1	Check visually and manually all cards tightness, locking, all sub-D connection tightness, all optic fibre cable tightness for proper layout. Check tightness of cubical doors safety locks.	OK		
2	Check the operation of all fans for free movements.	OK		
3	Check the tightness of connections of control cards. (IC Sch.)	—		
H	FIRE DETECTION UNIT			
1	Calibrate the FDU and set voltage in between: - Troxex make: 0 to ± 100 mV - Siemens Cerberus make: 0 to ± 50 mV			
2	Check and clean FDU pipe line by vacuum cleaner. (IC Sch.)			
3	Ensure the intactness / firmness of cable connection of Sicentocouplers of differential amplifiers provided below SR oil pump. (IC Sch.)			

Remarks if any:

पर्यवेक्षक के हस्ताक्षर Signature of JE/SSE



पश्चिम रेलवे
Western Railway

LOCOMOTIVE SHED, SABARMATI ELECTRICAL

[3-फेज विद्युत लोकोमोटिव आईए/आईबी/आईसी अनुसूची [निरीक्षण]
IA/IB/IC Schedule of 3-Phase Electric Locomotive [Inspection]



लोको संख्या Loco No. 30502 शिड्यूल Sch. IB शिड्यूल में लेने का दिनांक Date taken under Sch. 14-4-26

क्र.सं. SN	कार्यवाही कार्य/निरीक्षण का विवरण / (शिड्यूल) Detail of Work/ Inspection (Schedule)	कार्रवाई की गयी Action Taken	टीसीएन का नाम Name of TCN	हस्ताक्षर / टिप्पणी Sign / Remarks
A	ड्राइविंग डेस्क (Driving Desk)			
1	Clean the surfaces of the driver's desk with a soap solution and cloth, taking care not to allow water to enter electrical cubicles.	OK		
2	Clean the dust filter on the buzzer using a soft brush without using the pressure / compressed air.	OK		
3	Ensure that transparent rubber caps provided over the push buttons of driver desk.	OK		
4	Clean the battery, OHE and TE/BE meters in both cabs and check for zero error	OK		
5	Check tightness/operation/cable rubbing of connector, earthing of angle transmitter, all rubber gasket and all equipment located on BL and its box frame inside the panel. Clean contacts, check cap and its fitment tightness.	OK		
6	Check tightness of sub D located on BL.	OK		
7	Check ZPT switch located on pneumatic panel.	OK		
8	Check all earthing points located on equipment.	OK		
	LIGHTING AND AIRFLOW	OK		
B	हेड लाइट, कैब व कोरिडोर लाइट, फ्लैशर लाइट, मार्कर लाइट HEAD LIGHT, CAB & CORIDOR LIGHT, FLASHER LIGHT, MARKER LIGHT			
1	Check lighting units for physical damage, tightness of fixing connections, particularly look for signs of overheating or burning. Replace bulbs/tubes, which have failed.	OK		
2	Clean all internal light unit diffusers, bulbs or tubes and reflectors.	OK		
3	Clean the glasses of the headlight lamp and marker light.	OK		
4	Ensure that there is no entry of water through head light lamps.	OK		
5	Check for crack or breakage of the aluminum casing of headlight beams.	OK		
6	Check the proper operation of flasher light, marker light (red & white), headlight, cab lights, disc light, m/c. room light.	OK		
7	holder and clean glass Check headlight & reflector. check tightness of headlight convertor, Ensure sealing	OK		
8	Clean marker light glass and check tightness of connection. Check looseness, flashing in flasher unit.	OK		

Signature
15/4

क्र.सं. SN	कार्यवाही कार्य / निरीक्षण का विवरण / (शिड्यूल) Detail of Work/ Inspection (Schedule)	कार्रवाई की गयी Action Taken	टीसीएन का नाम Name of TCN	हस्ताक्षर / टिप्पणी Sign / Remarks
C	SB1 & 2, HB1 & 2 AND FB PANEL / CUBICLE			
1	Check the mounting block of 8.1 and 8.2 contactors by opening side windows and check the main contactor, pre charging contactor foundation bolt tips for looseness and overheating by removing arc-chutes from front side.	ok		
2	Check the condition of capacitors for leakage and check earth fault relay, resistance and discharging resistance for overheating. Check the tightness of all fastener lug, LEM sensor and CTs.	ok		
3	Ensure that the various foundations bolts, body support bolts and supports plates of FB cubicle are intact and tight. Check tightness of cubical doors safety locks.	ok		
4	Ensure tightness of various electrical connections, CBS in SB, HB cubicles.	ok		
5	Check back side of HB cubicle all SB connections for tightness/overheating/cable rubbing.	ok		
6	IC Sch.: a) Clean various equipment fitted inside the SB, HB & FB cubicle.	ok		
	b) Visual examination of various EPCs & EMCs fitted inside FB cubicle and ensure that there is no crackness of cover cylinder, leakage and they are functioning properly.	ok		
	c) Check the tightness of various PCB cards in central electronics.	ok		
	d) Check the RC input filter circuit for any abnormality and measure the resistance value.	ok		
	e) Take out the arc-chutes of pre-charging contactors for looseness of strips and ensure tightness of its armature locking nut by hand.	ok		
	f) Check the FB cable junction for tightness, cable rubbing and overheating.	ok		
D	HB CUBICLE :			
1	Check all equipments' fitment, tightness, intactness of base foundation bolts & body support.	ok		
2	Ensure earthing point/shunts of all panels/cubicles.	ok		
3	Check all MCB cable connections & Siemen connector tightness and no overheating.	ok		
4	Check tightness of 80 A & 150 A contactors connection and condition of tips.	ok		
5	Check 1000 V/415 V/110 V transformer connections for overheating and tightness of foundation bolts and checking of fuses for healthy connection.	ok		
6	Check tightness and condition of capacitor, LEM sensor below HB panel, earth fault relay & resistance connection, RLC plate, temperature sensor and all diode boxes.	ok		
7	Checking of OCB cable in HB 1 & 2 for overheating cable no.1121, 1122, 1123 (A & B).	ok		

30.06.2020
M.P.

हस्ताक्षर /
दिप्पणी

क्र.सं. SN	कार्यवाही कार्य / निरीक्षण का विवरण / (शिड्यूल) Detail of Work/ Inspection (Schedule)	कार्रवाई की गयी Action Taken	टीसीएन का नाम Name of TCN	हस्ताक्षर / दिप्पणी Sign / Remarks
E	SB CUBICLE:			
1	Check fasten lug connected on various equipment. Fitment, tightness.	ok		
2	Check intactness of base foundation bolts & body support. Check tightness of cubical doors safety locks.	ok		
3	Check all EMC connector lug position/layout. It should not be touch with plunger.	ok		
4	Visually check the connections on backside of SB 1 & 2 panel for overheating of WAGO connector etc.	ok		
5	Check tightness of DC-DC converter 24V/48V and Siemen connector for overheating.	ok		
6	Visually check all pressure switches, tightness of MCB, MCR relay, no voltage relay connections.	ok		
7	Key interlocking: Check all keys are present & the complete system is operational.	ok		
8	Ensure the healthiness; Visually check all pre/proper functioning of rotating switch 152, 154 and 237.1 on its various locations.	ok		
9	Check visually for any abnormality in SB, HB & FB cubicle	ok		
10	Ensure proper functioning of thermostat of central electronics. (IC Sch.)	ok		
F	MAIN ASSEMBLY BOX-1 & BOX-2:			
1	Clean the dust of enclosure using vacuum cleaner.	ok		
2	Check the metallic strip provided for spring action in the contact is intact. Replace the contact if required.	ok		
3	Check the connections for 25, 50 & 70 Sq. mm cables for their tightness. (IC Sch.)	ok		
4	Tightness of knife contact connection (female), 3-phase choke connection, Power contactor connection. (IC Sch.)	ok		

क्र.सं. SN	कार्यवाही कार्य / निरीक्षण का विवरण / (शिड्यूल) Detail of Work/ Inspection (Schedule)	कार्रवाई की गयी Action Taken	टीसीएन का नाम Name of TCN	हस्ताक्षर / दिप्पणी Sign / Remarks
G	CENTRAL ELECTRONIC RACK			
1	Check visually and manually all cards tightness, locking, all sub-D connection tightness, all optic fibre cable tightness for proper layout. Check tightness of cubical doors safety locks.	ok		
2	Check the operation of all fans for free movements.	ok		
3	Check the tightness of connections of control cards. (IC Sch.)	ok		
H	FIRE DETECTION UNIT			
1	Calibrate the FDU and set voltage in between: - Troxex make: 0 to ± 100 mV - Siemens/Cerberus make: 0 to ± 50 mV	ok		
2	Check and clean FDU pipe line by vacuum cleaner. (IC Sch.)	ok		
3	Ensure the intactness / firmness of cable connection of Sicent couplers of differential amplifiers provided below SR oil pump. (IC Sch.)	ok		

Remarks if any:

पर्यवेक्षक के हस्ताक्षर Signature of JE/SSE



पश्चिम रेलवे
Western Railway

LOCOMOTIVE SHED, SABARMATI ELECTRICAL

[3-फेज विद्युत लोकोमोटिव आईए/आईबी/आईसी अनुसूची [निरीक्षण]
IA/IB/IC Schedule of 3-Phase Electric Locomotive [Inspection]



लोको संख्या Loco No. 80802 शिफ्ट Sch. IR शिफ्ट में लेने का दिनांक Date taken under Sch. 18/4/18

A	PROPULSION SYSTEM: TRACTION CONVERTER			
1	Check the cable connection of Sichem connectors of differential amplifier provided below SR oil pump			
2	Check the condition of silica gel. If pink replace with dried silicagel			
	IC Sch :			
	a) Clean resonance circuit capacitor.			
	b) Cleaning by Vacuum cleaner			
	c) Clean DC link capacitors by blowing after opening side cover. Check that there is no sign of leakage from the capacitors			
3	d) Clean the isolating blade and spring contact of earthing switch. Lightly lubricate with the specified grease			
	e) Check the tightness of SR electronics PCB			
	f) Ensure sealing of incoming cable gland of control card rack to avoid dust entry			
	g) Check the ventilation fans of traction converters			
B	AUXILIARY CONVERTER 1, 2 & 3			
1	Visually check the connection of choke and transformer for overheating			
2	Check the tightness of foundation bolts and side supports for crackness. Check tightness of cubical doors safety lock			
3	Clean all dirt and dust deposit. Cleaning is by means of blowing out bushing off with a non-metallic brush or by rubbing with a cloth			
4	Check the tightness of sicem coupler and earthing point			
5	Check all wirings are secured and insulations are not damaged, burnt or eroded. Visually inspect all components for physical damage.			
6	Auxiliary rectifier & Inverter (CG Module & WRE module): check the security of bolted terminals and mechanical mounting of larger components.			
7	(IC Sch) Check the contacts of all modules and ensure proper tightness			

Approved by Media

क्र.सं SN	कार्यवाही कार्य / निरीक्षण का विवरण / (शिड्यूल) Detail of Work/ Inspection (schedule)	की गयी कार्यवाही Action Taken	टीसीएन का नाम Name of TCN	हस्ताक्षर / टिप्पणी Sign/ Remarks
C	TRACTION CONVERTER (SR)			
1	Examine all electrical equipment of traction converter for signs of dirt, corrosion, damage etc. Remove all dust/dirt deposits from the connection insulators. Check the tightness of foundation supports and side supports for crackness. Check tightness of cubical doors safety locks			
2	Check the main contactor, pre-charging contactor foundation bolt and tips for looseness and overheating by removing archutes. Check and ensure tightness of pre-charging contactor armature locking nut and strips			
3	Check all connected fast on lug, bus bar connection, CTs, all LEM sensor connections, gate unit connections, pre-charging, MUB and its earth fault resistance, DC link and series resonance capacitor for leakage, tightness and overheating			
4	Check the connection of earth fault relay. DC link sensor and all the earthing point of SR1, SR2			
5	Check BDV of traction converter oil (in GTO based loco), should be 30KV			
6	Carry out DGA of traction converter oil by C&M Lab (in GTO base loco)			
7	Check the oil level indicator situated on the conservator(Expansion Tank). If the oil is below the minimum mark-top up with the specified oil. Check for any signs of leakage, Range Oil level middle strip +/- 1.5"			
8	Check for any leakage, loose, missing of screw from oil pipe joint, two flange joints, wire braided pipe, valve set vent plug, stoochi coupling, top bottom oil pipe, back body of valve set			
9	SR-1 & SR-2 terminal, if difference of inductance between any two phase is < 0.005mH, allow the motor for next schedule			
10	If the value is more than 0.005 mH, measure the inductance of individual TM and if difference between two phase of motor is < 0.015mH, allow the motor for next schedule. If > 0.015mH, remove the motor			
11	Check the cable connection of Sichem connectors of differential amplifier provided below SR Oil pump.			
12	Check ventilation fan of Traction converter.			

Remarks if any:

Inductance Value

1:-	<u>UV</u> 4.028	<u>UW</u> 4.029	<u>VW</u> 4.032
2:-	4.337	4.342	4.346
3:-	4.035	4.040	4.039
4:-	4.442	4.446	4.440
5:-	4.376	4.384	4.382
6:-	4.540	4.539	4.552

पर्यवेक्षक के हस्ताक्षर Signature of JE/SSE



पश्चिम रेलवे
Western Railway

LOCOMOTIVE SHED, SABARMATI ELECTRICAL

[3-फेज विद्युत लोकोमोटिव आईए/आईबी/आईसी अनुसूची [निरीक्षण]
IA/IB/IC Schedule of 3-Phase Electric Locomotive [Inspection]

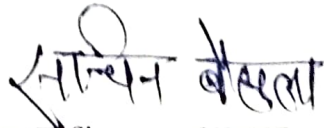


लोको संख्या Loco No. 20502 शिड्यूल Sch. IB शिड्यूल में लेने का दिनांक Date taken under Sch. 17/4/26

AUXILIARY MACHINES SCHEDULE

क्र.सं. SN	कार्यवाही कार्य / निरीक्षण का विवरण / (शिड्यूल) Detail of Work/ Inspection (schedule)	की गयी कार्यवाही Action Taken	टीसीएन का नाम Name of TCN	हस्ताक्षर/ टिप्पणी Sign/ Remarks
A	Oil circulating pumps (SR & TFP)			
	1) Visually examine all SR&TFP oil pumps (Four nos.) for any oil leakage / any abnormal sound and take needful action.	oh		
	2) Check the mechanical support fasteners of all four oil pumps.	oh		
	3) Check and secure rubbing of SR/TFP Cable.	oh		
B	During schedule check all Auxiliary Blower and Motors: -	oh		
	1) Check tightness of foundation bolts of all auxiliary motors.	oh		
	2) Visual examination / RDPT of impeller of OCB.	oh		
	3) Check the condition of all flexible hoses connected with various auxiliary blowers and replace, if required.	oh		
	4) Check tightness of SB and cable connections at terminal box.	oh		
	5) Ensure that the terminal covers of various auxiliary motors are intact and water tight. Special attention needs to be given to main compressor, OCB, TMB motor cable layout and water tightness of its entry point into the motor terminal.	oh		
	6) Clean commutator of CPA motor. Check carbon brush and its holding assembly and replace worn out / defective carbon brushes and associated equipments.	oh		
	7) Check availability of earthing shunt.	oh		
	8) Check condition of cab fan foundation, secure rubbing of wire, and check freeness of fan blade.	oh		
C	IC Sch.: Besides of the above Sch. following attention needs to be given: -			
	1) Check the electrical connections of all four oil pumps.			
	2) Check the electrical connections of all auxiliary motors.			
	3) Greasing of all motors with recommended grease or RR-3 Servo Gem Grease ,6 to 7 Stroke ,15 gm.			

Remarks if any:


पर्यवेक्षक के हस्ताक्षर Signature of JE/SSE



पश्चिम रेलवे
Western Railway

LOCOMOTIVE SHED, SABARMATI
ELECTRICAL

[3-फेज विद्युत लोकोमोटिव आईए/आईबी/आईसी अनुसूची [निरीक्षण]
IA/IB/IC Schedule of 3-Phase Electric Locomotive [Inspection]



लोको संख्या Loco No. 35502 शिड्यूल Sch. IR शिड्यूल में लेने का दिनांक Date taken under Sch. 17.4.26

ट्रैक्शन मोटर शिड्यूल **TRACTION MOTOR SCHEDULE**

(A) Sch. IA/IB/IC: TM Schedule

क्र.सं. SN	कार्यवाही कार्य / निरीक्षण का विवरण / (शिड्यूल) Detail of Work/ Inspection (schedule)	की गयी कार्यवाही Action Taken	टीसीएन का नाम Name of TCN	हस्ताक्षर / टिप्पणी Sign/ Remarks
1	Examine all traction motors for signs of damage, dents or other defects caused by ballast. Check air outlets are not obstructed in any way.	checked	Ajay Kumar	Santosh
2	Check traction power cables, speed sensor and temperature sensor cables and ensure that they are not chamfered or damaged in any way.			
3	Check Tightness of both end cover bolts of Traction Motor.			
4	Check the intactness of the junction box cover and bolt			
5	Open TM Junction box at body side and check the tightness of connections			
6	Check the condition of bellows and replace if required.			

(B) Sch. IC: TM Schedule & Greasing

क्र.सं. SN	कार्यवाही कार्य / निरीक्षण का विवरण / (शिड्यूल) Detail of Work/ Inspection (schedule)	की गयी कार्यवाही Action Taken	टीसीएन का नाम Name of TCN	हस्ताक्षर / टिप्पणी Sign/ Remarks
1	Check for intactness of Grease nipple on NDE & DE side. Lubricate the both end bearings using a grease gun with specified grease and send old grease sample to Lab for testing metal content.	checked	Ajay Kumar	Santosh
2	Check the condition of bearing by seeing the grease, which comes out from grease inlet. Check the condition of grease too.			
3	Send grease sample to Lab for testing metal content, whenever grease available.			
4	Check under truck TM cables for any rubbing against metallic body and secure cables properly.			
5	If applicable following to be check:			
	i) Ensure tightness of all bolts of all Hurth coupling.			
	ii) Check oil level of Hutch coupling and send sample of oil to Laboratory.			
	iii) Check all TM coupling membrane for crack and oil leakages.			
	iv) Clean sleeve and star of Hurth coupling.			
	v) Check and replace bolt if found defective treads of bolts of sleeve and star of Hutch coupling.			
	vi) Check free movement of Hutch coupling.			

Remarks if any:

सन्तोष वैद्य



पश्चिम रेलवे
Western Railway

LOCOMOTIVE SHED, SABARMATI ELECTRICAL

IA/IB/IC Schedule form for KAVACH (TICAS) in WAG9HC Locomotive



Loco no: 30502

Schedule: IB

Date: 18/04/26

Sr.	Description of the Equipment	Parameter for Quality check	Details of Work Done		Remarks	IA/IB	IC
			Cab#1	Cab#2			
1.1.1	LPOCIP	Cleanliness	OK	OK		✓	✓
1.1.2		Physical condition	OK	OK		✓	✓
1.1.3		Fixing condition	OK	OK		✓	✓
1.2.1	Cab Radio	Cleanliness	OK	OK		✓	✓
1.2.2	Display Unit (If applicable)	Physical condition	OK	OK		✓	✓
1.2.3		Fixing condition	OK	OK		✓	✓
2.1.1	Buzzer	Cleanliness	OK	OK		✓	✓
2.1.2		Physical condition	OK	OK		✓	✓
3.1.1	Radio Unit	Cleanliness	OK	OK		✓	✓
3.1.2		Physical condition	OK	OK		✓	✓
3.1.3		Intactness of connectors and couplers	OK	OK		✓	✓
3.1.4		Fixing condition	OK	OK		✓	✓
4.1.1	LTE Router-1 (If applicable)	Cleanliness	OK	OK		✓	✓
4.1.2		Physical condition	OK	OK		✓	✓
4.1.3		Intactness of connectors and couplers	OK	OK		✓	✓
4.1.4		Fixing condition	OK	OK		✓	✓
5.1.1	LTE Router-2 (If applicable)	Cleanliness	OK	OK		✓	✓
5.1.2		Physical condition	OK	OK		✓	✓
5.1.3		Intactness of connectors and couplers	OK	OK		✓	✓
5.1.4		Fixing condition	OK	OK		✓	✓
7.1.1	RFID Reader	CABLE Conduit clamping condition	OK	OK		✓	✓
7.1.2		RFID Reader Mounting fixture welding Intactness (Visual check)	OK	OK		✓	✓
7.1.3		Ensure intactness of welding by Magnetic particle testing/DPT	OK	OK			✓
7.1.4		Availability and intactness of Sling for RFID Reader	OK	OK		✓	✓
7.1.5		Intactness of Guard for RFID reader.	OK	OK		✓	✓
8.1.1	Onboard KAVACH	Cleanliness	OK	OK		✓	✓
8.1.2		Physical condition	OK	OK		✓	✓
8.1.3		Intactness of connectors and couplers	OK	OK		✓	✓
8.1.4		Fixing condition	OK	OK		✓	✓
8.1.5		Feeding/ensuring of correct wheel dia. As per the wheel dia. Measurement carried out by Shed.	OK	OK		✓	✓
8.1.6		Remove connectors/couplers	OK	OK		✓	✓
8.2.1	Counter Readings	Note EB counter reading	OK	OK		✓	✓
8.2.2		Note Isolation counter reading	OK	OK		✓	✓
8.2.3		Note SOS counter reading	OK	OK		✓	✓
8.3.1	Can Input Box (If applicable)	Cleanliness	OK	OK		✓	✓
8.3.2		Physical condition	OK	OK		✓	✓
8.3.3		Intactness of connectors and couplers	OK	OK		✓	✓
8.3.4		Fixing condition	OK	OK		✓	✓

Sr.	Description of the Equipment	Parameter to be checked	Details of work done (Only one item / loco)	Remarks	IA/IB	IC
8.4.1	E70/CCB	Cleanliness	OK		✓	✓
8.4.2	Interface Unit (If applicable) (HBL)	Physical condition	OK		✓	✓
8.4.3		Intactness of connectors and couplers	OK		✓	✓
8.4.4		Fixing condition	OK		✓	✓
10.1.1	Isolation Box	If isolation box kept in Isolation position whether LPOCIP is indication the status or not	OK		✓	✓
10.1.2		Cleanliness	OK		✓	✓
10.1.3		Physical condition	OK		✓	✓
11.1.1	Addl. Q51 (HBL)	Cleanliness	OK		✓	✓
11.1.2		Physical Condition	OK		✓	✓
11.1.3		Connection/Contacts	OK		✓	✓

Under Frame

Sr.	Description of the Equipment	Parameter to be checked	Details of work done (Only one item / loco)	Remarks	IA/IB	IC
12.1.1	Speed Sensor-1	Visual Check of speed sensor cables	OK		✓	✓
12.1.2		Physical condition of PG mounting and Junction Box	OK		✓	✓
12.1.3		Remove the speed sensor and clean any excess grease	OK			✓
12.2.1	Speed Sensor-II	Visual Check of speed sensor cables	OK		✓	✓
12.2.2		Physical condition of PG mounting and Junction Box	OK		✓	✓
12.2.3		Remove the speed sensor and clean any excess grease	OK			✓

Roof Area

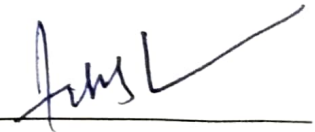
Sr.	Description of the Equipment	Parameter to be checked	Details of work done (Only one item / loco)	Remarks	IA/IB	IC
13.1.1	GSM/GNSS (GPS)	Physical condition	OK		✓	✓
13.1.2	Antenna of roof (If some makes only one unit available)	Cable conduit clamping condition	OK		✓	✓

Sr.	Description of the Equipment	Parameter to be checked	Antenna#1	Antenna#2	IA/IB	IC
14.1.1	Antenna on Roof of Cab#1	Physical condition	OK		✓	✓
14.1.2		Cable conduit clamping condition	OK		✓	✓
14.1.3		Checking Reverse Power Status	OK		✓	✓
14.2.1	Antenna on Roof of Cab#2	Physical condition	OK		✓	✓
14.2.2		Cable conduit clamping condition	OK		✓	✓
14.2.3		Checking Reverse Power Status	OK		✓	✓
14.3.1	LTE Antenna-1 (If applicable)	Physical condition	OK		✓	✓
14.3.2		Cable conduit clamping condition	OK		✓	✓
14.4.1	LTE Antenna-2 (If applicable)	Physical condition	OK		✓	✓
14.4.2		Cable conduit clamping condition	OK		✓	✓

Sign & Name of TCN:



Sign & Name of JE/SSE:





पश्चिम रेलवे
Western Railway

LOCOMOTIVE SHED, SABARMATI
ELECTRICAL

[3-फेज विद्युत लोकोमोटिव एच एल सी अनुसूची [निरीक्षण]
HLC Schedule of 3-Phase Electric Locomotive [Inspection]



लोको संख्या Loco No. **30502**

शिड्यूल Sch. **IB**

दिनांक Date

1.0 Schedule for maintenance and activities for Hotel Load Converter:

Activities in IA/IB/IC schedules:

Sr.	Activities	IA/IB	IC	Deficiency Noticed	Action Taken
1.	Check functioning of the blowers and churning fans. Replace in case of any abnormality. Check direction if required.	✓	✓	Check ok	
2.	Inspect the contact surfaces of main/pre-charge/auxiliary contactors for any sign of wear, pitting or erosion. If the contacts are significantly worn, consider replacing them.	✓	✓	OK	
3.	Clean all Suction Filters of hotel load converter.	✓	✓	OK	
4.	After schedule of inspection, erase FDP memory.	✓	✓	OK	
5.	Check parameters of HLC in working conditions. - Output voltage: 750V +5%/-2%; - Line to Earth voltage peak : <800V; - O/p Voltage THD: < 10%; - I/p current THD (full load): <5% (up to 20th harmonics); - O/p voltage unbalance: >3%.	751V 435V, 436V, 434V — — —	✓	OK	
6.	Check the sealing of the cable entry into the panel (to stop the dust entry).	✓	✓	OK	
7.	Check the earth connections of the HLC and Output contactor panels.	✓	✓	OK	
8.	Check the tightness/intactness of connectors of the panel.	✓	✓	OK	
9.	Check the door locks of the panel.	✓	✓	OK	
10.	Check the working of display unit of HLCs & functioning of keypad keys.	✓	✓	OK	
11.	Visual checking of blower transformer, if available.	✓	✓	OK	
12.	Cleaning of front door filters, if provided.	✓	✓	OK	
13.	Check both HLCs on load arrangement for ensuring proper working on line. Run HLC on load for 5-10 minutes. Note current and voltage. Check all indication at control box is proper or not.	✓	✓	OK	
14.	Check the condition of blocking diode fitted in SB-2 panel as per RDSO MS-488.	✓	✓	OK	

Sr.	Activities	IA/IB	IC	Deficiency Noticed	Action Taken
15.	Ensure proper working of HOG ON Command Indication lamp provided in D- Panel as per RDSO MS-489.	√	√	OK	
16.	Check working of fault reset command with the help of control box.	√	√	OK	
17.	Check air flow at suction filter and exhaust louvers position for ensuring proper working of blower fans. Average suction air flow should be >2.5 m/s for Medha & BHEL make and >3.5 m/s for Siemens & AAL make HLCs. The measurement should be taken at the points mentioned in Figure-1 below:	√	√	OK	

Additional activity in IC schedule

Sr.	Activities	IA/IB	IC	Deficiency Noticed	Action Taken
1.	Check door gaskets for damages. If found damaged, change the same.	X	√	—	
2.	Check the dust ingress into the HLC panel and output contactor panel. Use a vacuum cleaner to clean.	X	√	—	
3.	Check the cable tie wraps of the power cables. Check for sign of overheating.	X	√	—	
4.	Measure the overall capacitance of each DC link and replace if needed. (12mF ± 5% – Siemens & AAL, 10.8 mF ± 5% – Medha, 10.4mF ± 5% – BHEL). Ensure proper tightening torque of fasteners.	X	√	—	
5.	Check tightness of connections at input, output and control cables.	X	√	—	
6.	Check the power connection for color changes/overheating and change the cables and sockets, if required.	X	√	—	
7.	Check tightness of connections of power cables.	X	√	—	
8.	Check the pre-charging contactors for any signs of physical damage, overheating or discoloration. Check for loose connections and ensure that all parts are in good condition.	X	√	—	
9.	Check the contacts of the main contactors for any signs of physical damage, overheating or discoloration. Check for loose connections and ensure that all parts are in good condition.	X	√	—	
10.	Check the contacts of the auxiliary Contactors for any signs of physical damage, overheating or discoloration. Check for loose connections and ensure that all parts are in good condition.	X	√	—	
11.	Clean the heat sinks of HLCs.	X	√	—	

r.	Activities	IA/IB	IC	Deficiency Noticed	Action Taken
2.	Check the tightness of mounting / fixation bolts with the loco body.	X	√	—	
13.	Check the output contactors and o/p Power cable connection for any overheating physical damage or discoloration. Check for loose connections and ensure that all parts are in good condition	X	√	—	
14.	Visually check the sine filter capacitor and measure its value. Ensure proper tightening torque of fasteners.	X	√	—	
15.	Visually inspect the RC filter and discharge resistors for signs of overheating.	X	√	—	
16.	Visually inspect pre-charge resistor. Replace if found damaged or overheated.	X	√	—	
17.	Check phase sequence of all 04 IV couplers.	X	√	—	

A	B	C		J	K	L
D	E	F		M	N	O
G	H	I		P	Q	R

Figure – 1: Air flow measurement from HLC Suction Filters (Loco side wall)

- Fault log and driver Bookings

2.0 Maintenance of IV Couplers

Sr.	Activities	IA/IB	IC	Deficiency Noticed	Action Taken
1.	Check IV coupler cover/body for any damage and replace if required.	√	√	OK	
2.	Check visually for any sign of damage, flash mark and overheating. Repair if required.	√	√	OK	
3.	Check for intactness of sealing ring gasket on socket of locomotive.	√	√	OK	
4.	Clean IV coupler and lubricate moving parts if found sluggish.	√	√	OK	
5.	Check IV coupler Ratchet assembly and ensure proper locking/unlocking & ensure intactness of all Hardware (nuts/bolts/washers etc.).	√	√	OK	
6.	Ensure intactness of both HEX small bolt (bearing screw) and both HEX big bolt (Bearing Pivot Bolt).	√	√	OK	
7.	Ensure proper working of locking bolt.	√	√	OK	
8.	Ensure intactness of both Allen screw.	√	√	OK	
9.	Ensure intactness of insulated base foundation bolt. It should be straight to the holes given on the coupler edge.	√	√	OK	
10.	Ensure the proper condition of Lever tie bow Assembly gap. Check with "NO GO Gauge".	√	√	OK	
11.	Check IV coupler mobile contact pin (Phase & control) for flashing/free movement by pressing and replace if stuck up / sluggish.	√	√	OK	
12.	Clean coupler pins with suitable cleaners.	√	√	OK	
13.	Ensure that couplers are in closed conditions to prevent cover damage and ingress of water & dust.	√	√	OK	
14.	Check phase sequence at all 4 couplers.	√	√	OK	
15.	Check Insulation between Control Pins (C1 & C2) w.r.t. body of IVC at 500 VDC. This should be >10 M ohm	√	√	OK	
16.	Check Insulation between high voltage supply pins w.r.t. body of IVC with 1500 VDC Megger. This should be >100 M ohm.	√	√	OK	

Additional activity in IC schedule

Sr.	Activities	IA/IB	IC	Deficiency Noticed	Action Taken
17	Replace O ring (sealing gasket) of IV coupler.	X	√	—	

Maintenance of UIC Couplers

sr.	Activities	IA/IB/IC	Deficiency Noticed	Action Taken
1.	Both sides UIC coupler cover tension to be checked. If found not proper then replace the same.	√	OK	
2.	Ensure proper unlocking/locking of covers of both sides UIC couplers before and after inspection. All nut-bolts should be intact.	√	OK	
3.	Both sides UIC coupler female contact pin to be checked for oxidation/flashing, pin diameter and locking (to be done without opening of UIC coupler).	√	OK	
4.	UIC coupler must be in closed condition to preventing ingress of dust and water.	√	OK	
5.	In UIC coupler, condition of all red colour gaskets needs to be checked.	√	OK	
6.	During load / no load testing of HLCs, functionality of all concerned UIC pins needs to be checked.	√	OK	

Any other remarks:

Name of TCN: SANTOSH KUMAR

Sign of TCN: Santosh kr.

Name of Supervisor:

Sign of supervisor: Adesh Tyagi